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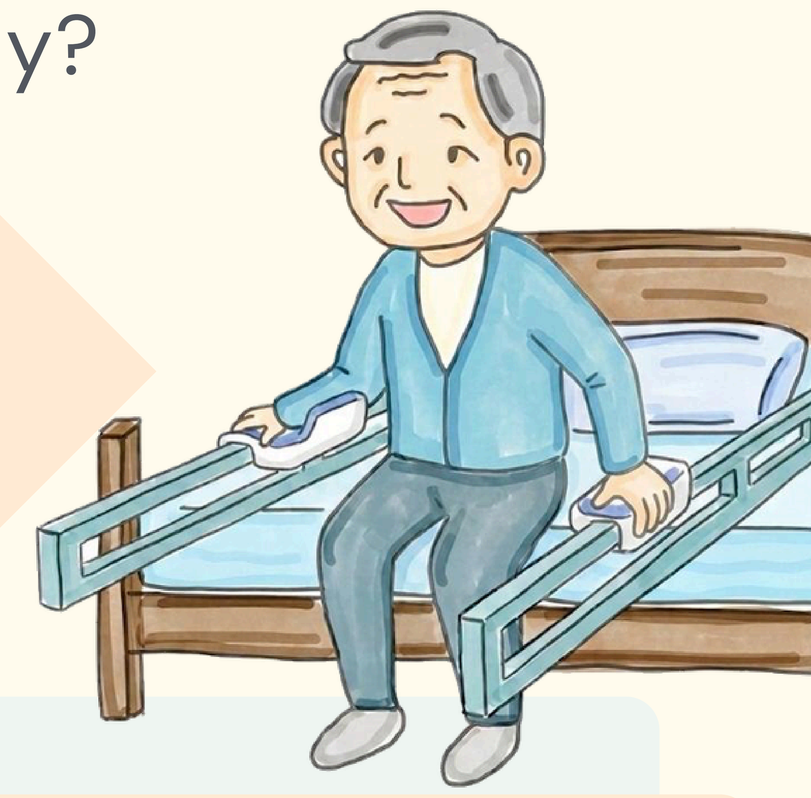
Alleviate

Problem Statement

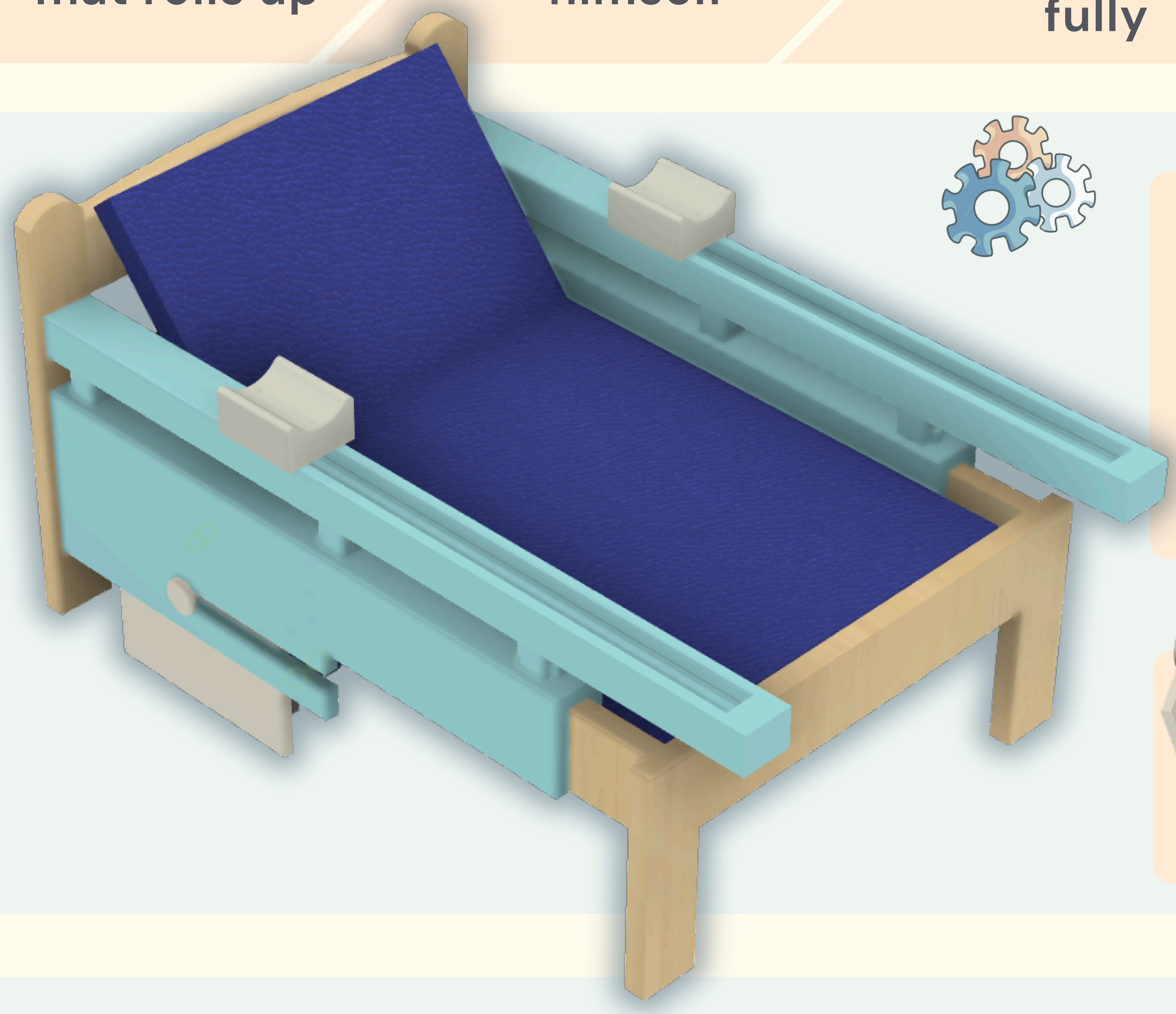
HMW enable **elderly** with **limited mobility** to safely transition between **sitting** on a bed and **standing** independently?

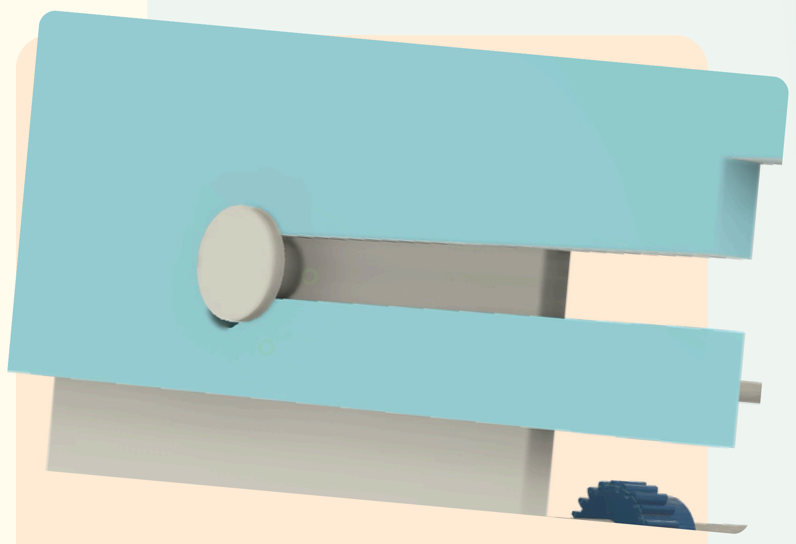


How It Works

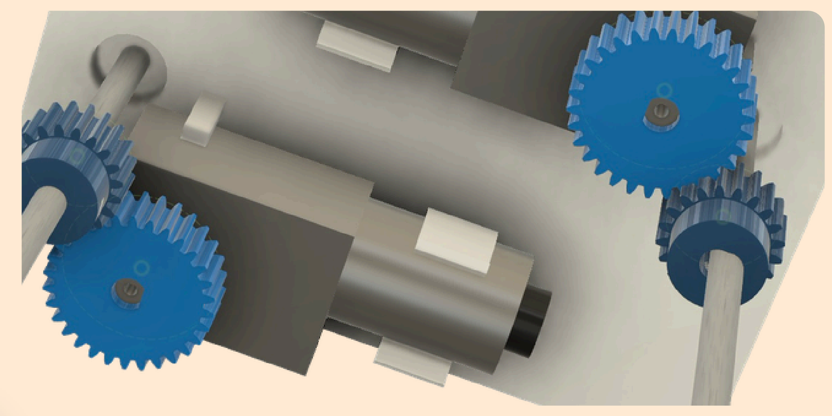
User presses a button → Armrests move forward, mat rolls up → Legs reach ground, user stabilises himself → Frame supports user until he stands up fully

Features

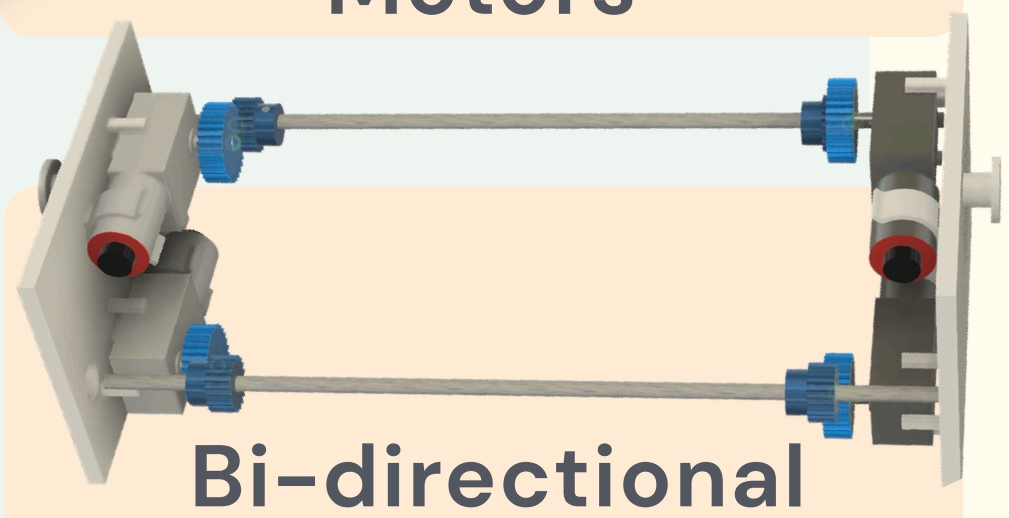




Ergonomic Integration; Detachable



Synchronised Motors

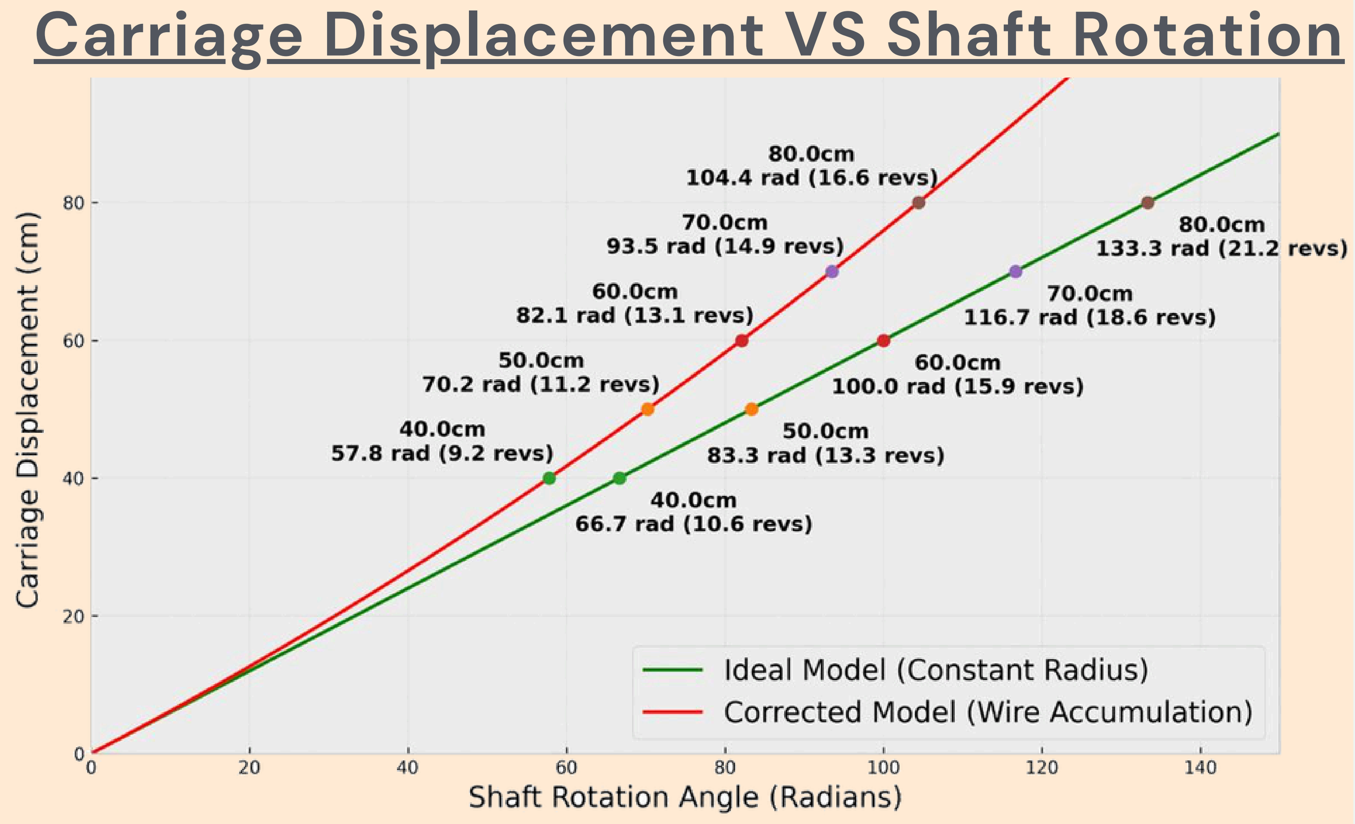


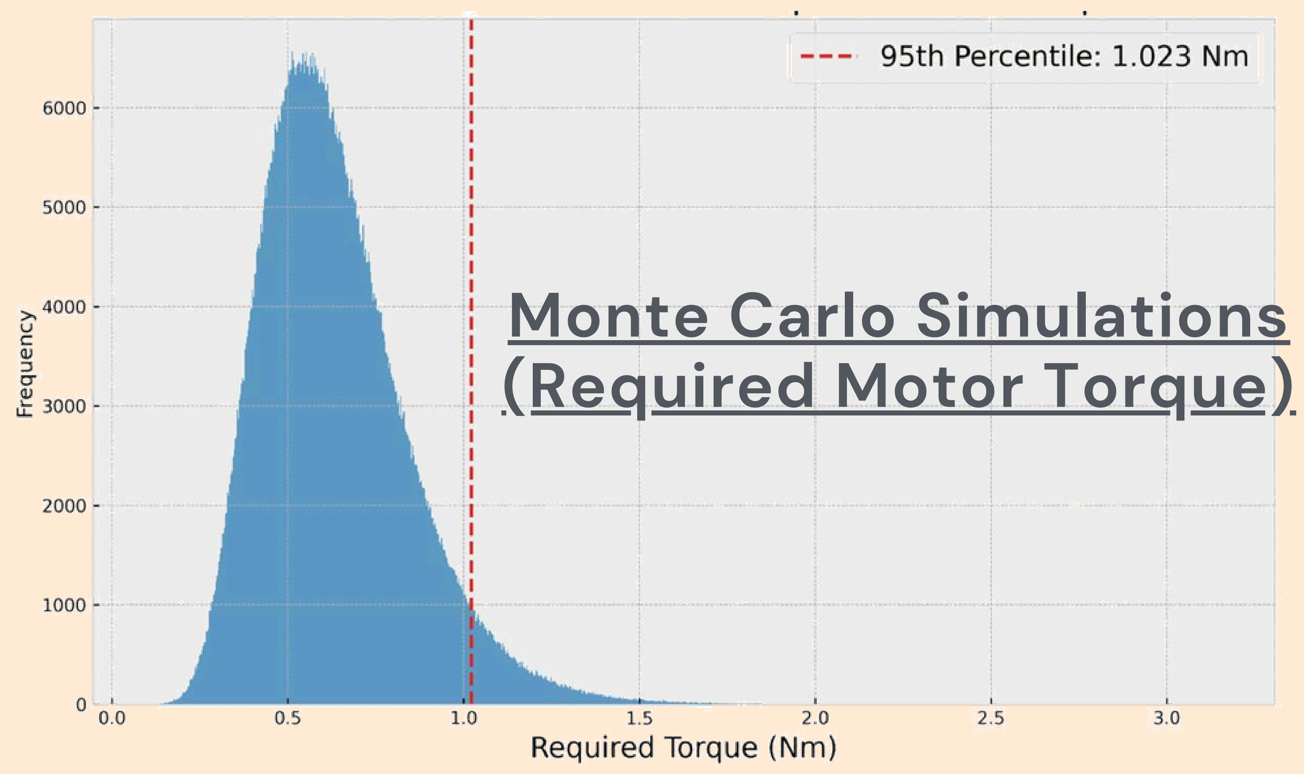
Bi-directional

Concepts Used

CaDDE

Carriage Displacement VS Shaft Rotation



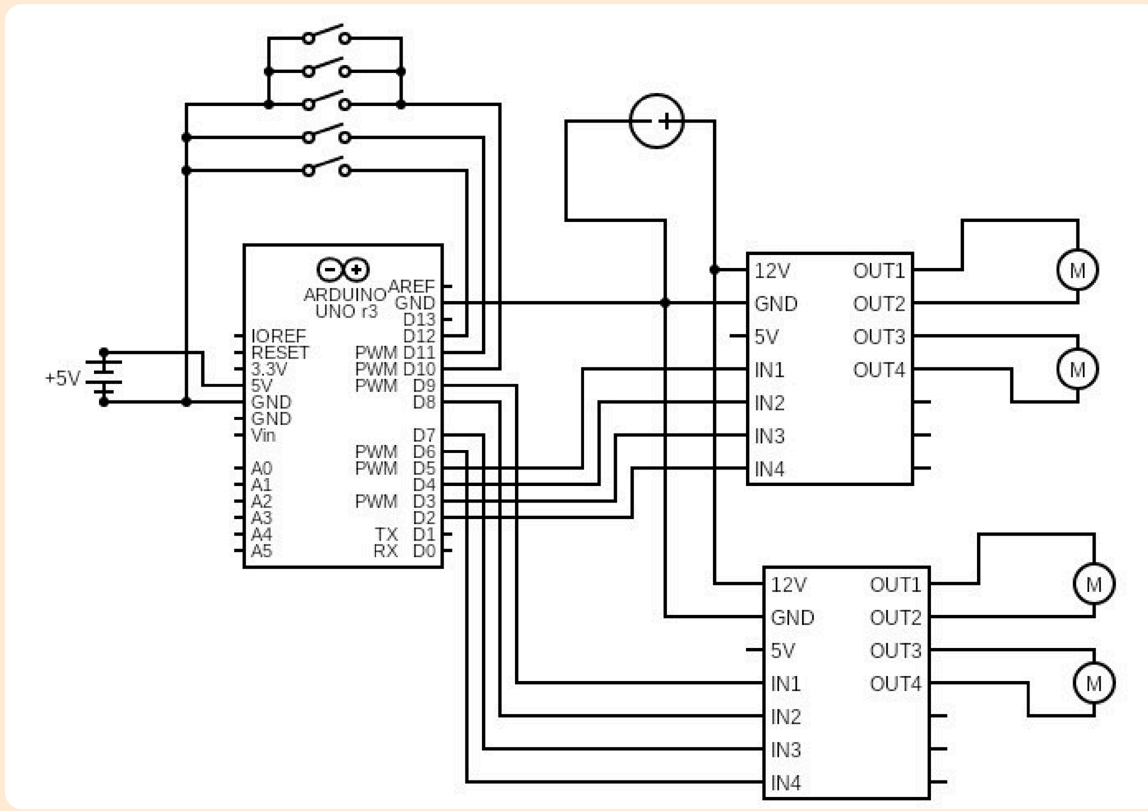


Monte Carlo Simulations (Required Motor Torque)

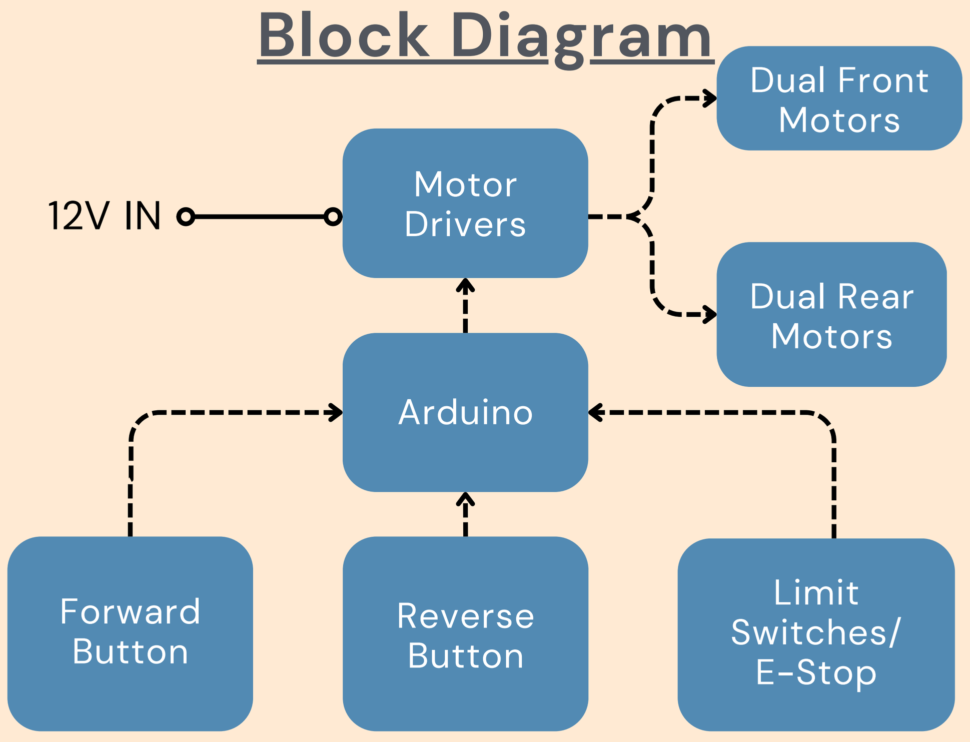
Led to the selection of a motor with more than **11.2 to 13.3** RPM for a **50cm** transfer (1 minute), and an operational torque greater than **1.023N** with a 2x **Factor Of Safety**

Circuits

Circuit Diagram



Block Diagram



Input pull-up logic to prevent floating states;
Motor drivers to drive bi-directional DC motors

Structures & Materials

Material Selection

Utilises aluminium 6061-T6 hollow rectangular sections (50x50x4 mm) for rails, selected for the **high strength-to-weight ratio** [Sy = 276 MPa, density 2700 kg/m³], over mild steel.

Non load-bearing connectors and housing are fabricated in 3D printed **PLA** for **compressive strength** and **rapid fitment**.

Beam Analysis

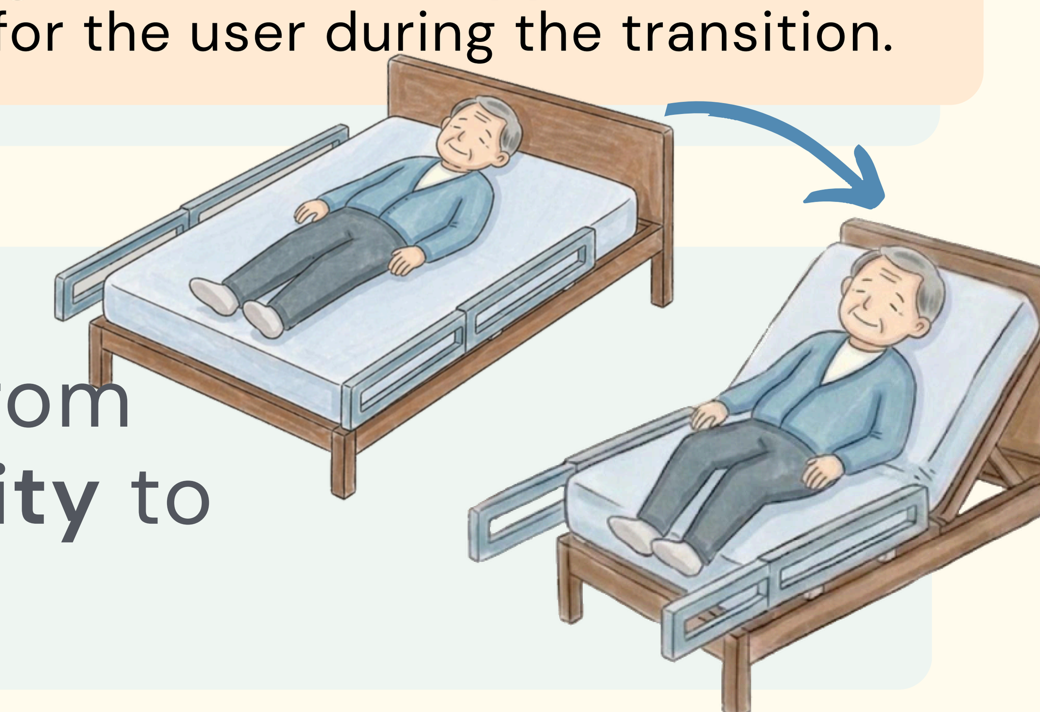
To ensure user safety, rail was modelled as a **fixed-free cantilever** at a maximum extension of 800 mm, under a 400 N load.

Analysis yielded a maximum **Von Mises stress** of 109.5 MPa, maintaining a **FOS** of 2.5.

Serviceability & Safety

The resulting **tip deflection** of 2.6 mm remains within the L/300 serviceability limit, thus ensuring a **rigid and stable support interface** for the user during the transition.

Future Improvements



Integrate a **back-rest articulation system**, transitions users from **supine** to **seated** position, providing **hospital-bed functionality** to standard bed frames, **reducing cost** of mobility care.